# Hello Colloquium

A demo presentation with math, code, and figures

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# What is Colloquium?

- Agent-native slide creation tool for research talks
- Markdown-based and git-friendly
- AI agents can drive it programmatically
- Single self-contained HTML output

# LaTeX Math Support

The loss function for RLHF with a KL penalty:

$$\mathcal{L}(\theta) = -\mathbb{E}_{x \sim D}\left[\log \sigma\left(r_\theta(x_w) - r_\theta(x_l)\right)\right]$$

Inline math works too: $\nabla_\theta J(\theta) = \mathbb{E}_{\pi_\theta}[R \cdot \nabla_\theta \log \pi_\theta(a|s)]$

# Code Highlighting

```python
from colloquium import Deck

deck = Deck(title="My Talk", author="Researcher")
deck.add_title_slide(subtitle="A research presentation")
deck.add_slide(
    title="Key Results",
    content="Our method achieves **state-of-the-art** performance.",
)
deck.build("output/")
```

# Two Equal Columns

- Point one
- Point two
- Point three

- Result A: **95.2%**
- Result B: **87.4%**
- Result C: **91.8%**

# Asymmetric Columns (60/40)

This wider column has the main explanation text. The 60/40 split gives more room to the primary content while keeping a sidebar for supplementary info.

- Key finding one
- Key finding two

> *"Supplementary details go in the narrower column."*

Supporting points:

1. Evidence A
2. Evidence B

# Centered Image

# Image with Text

Colloquium supports images in any layout. Here the wordmark sits in the wider column alongside explanatory text.

Images auto-scale to fit their container while maintaining aspect ratio.

# Key Results

# Experimental Results

| Model | Accuracy | F1 Score | Training Time |
|---|---|---|---|
| Baseline | 82.1% | 79.3% | 2h |
| Ours (small) | 91.4% | 89.7% | 4h |
| Ours (large) | **95.2%** | **93.8%** | 12h |

> *"The results demonstrate significant improvements across all metrics."*

# Centered & Large Text

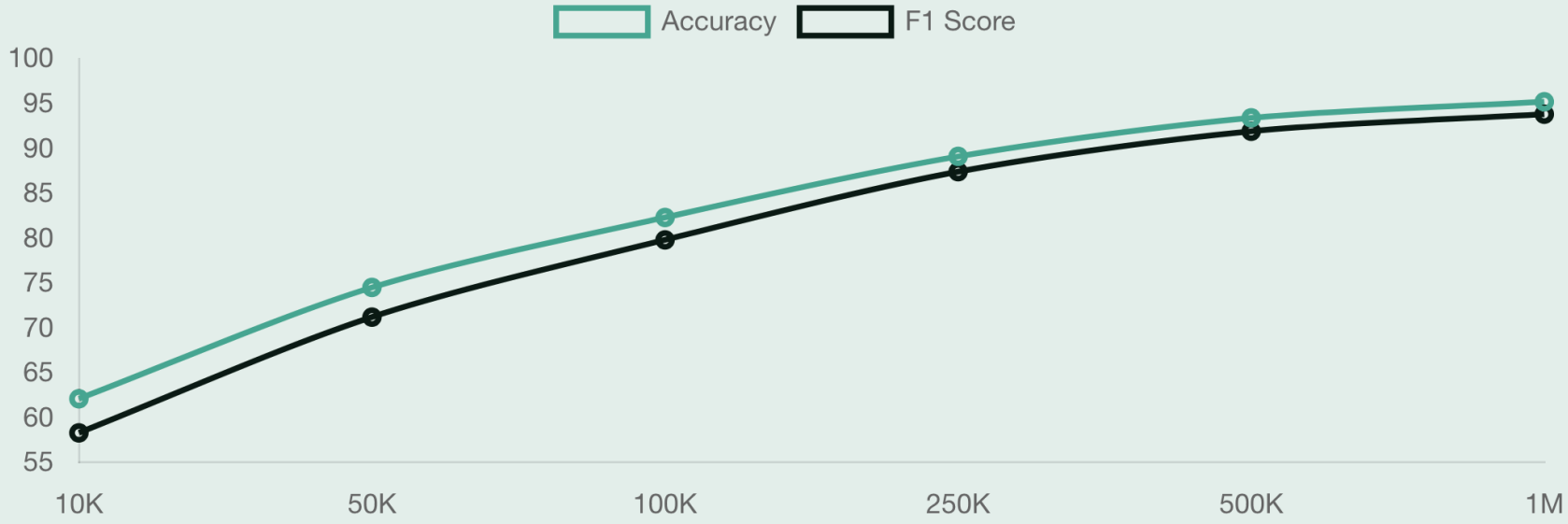
This slide uses the `align` and `size` directives to center all text and increase the font size.

Great for emphasis slides.

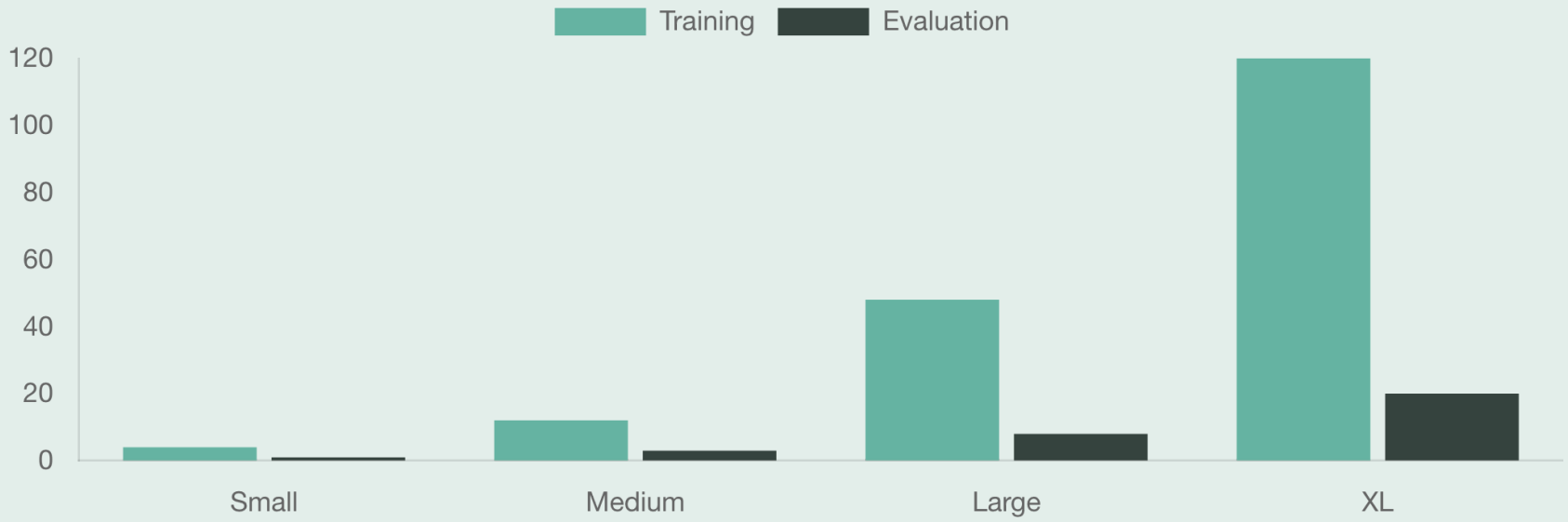
## Vertically Centered Title

All content on this slide is vertically centered, like a title slide but with `##` heading style.

# Training Performance



# Resource Usage



# Conclusions

1. Colloquium makes slide creation **fast** and **reproducible**
2. Full LaTeX math support for academic presentations
3. Git-friendly markdown source files
4. AI agents can create and modify presentations programmatically

# Thank You

Questions?

github.com/interconnects-ai/colloquium